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Jurassic magnetic stratigraphy from Umbrian (Italian) land sections

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The Umbrian Apennines were the site of pelagic sedimentation throughout most of the Jurassic. Magnetic stratigraphy from four sections spanning many of the Jurassic stages indicates that the geomagnetic field at this time was characterized by two intervals of mixed polarity, separated by an interval of predominantly normal polarity corresponding to the Jurassic "quiet-zone" in the oceanic magnetic anomaly record. Unfortunately, ammonites are poorly preserved or absent throughout most of these sections; the duration of this "quiet-interval" cannot be well defined, although it is probably restricted to the Callovian and Oxfordian stages.

1. Introduction

As the Mesozoic Tethys began to develop in the Lias, the carbonate platforms of central Italy were isolated on the southern continental margin of the new ocean. The Jurassic palaeogeography of Umbria is a consequence of continental margin subsidence of a large Triassic epicontinental carbonate platform [1]. An extensive pelagic realm developed characterized by basinal areas, and intervening seamounts with relatively condensed sedimentation [2]. The continental margin subsidence, which began at the Hettangian/Sinemurian boundary (Lower Lias) continued throughout the Mesozoic, giving rise to a sequence of pelagic limestones which constitute the Mesozoic of the Umbrian Apennines. A generalized stratigraphic column (Fig. 1) gives the formational names with approximate ages and stratigraphic thicknesses.

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The Calcare Massiccio of Hettangian age (Fig. 1) is overlain by a pelagic calcareous unit (Corniola Formation, middle Lias) which in turn is overlain by a more marly group of rocks (M. Serrone marls and Ammonitico Rosso, Upper Lias/Lower Dogger). During the Dogger, the marls give way to a cherty limestone referred to as the Calcari Diasprigni, the lower part of which is occasionally known as the "Calcari Bianchi con Selce" (Fig. 1). The Calcari Diasprigni is a white or grey-green limestone with abundant green chert. The amount of chert decreases towards the top of the Jurassic, and the Maiolica Formation, a white pelagic limestone, characterizes the Lower Cretaceous. The Paleocene and Cretaceous formations (Scaglia and Maiolica) have yielded well-defined magnetic stratigraphies which have been correlated with foraminiferal and nannofossil zonations (e.g. [3–5]). We present paleomagnetic results from four sections covering much of the Jurassic, from the top of the Hettangian to the Upper Tithonian.

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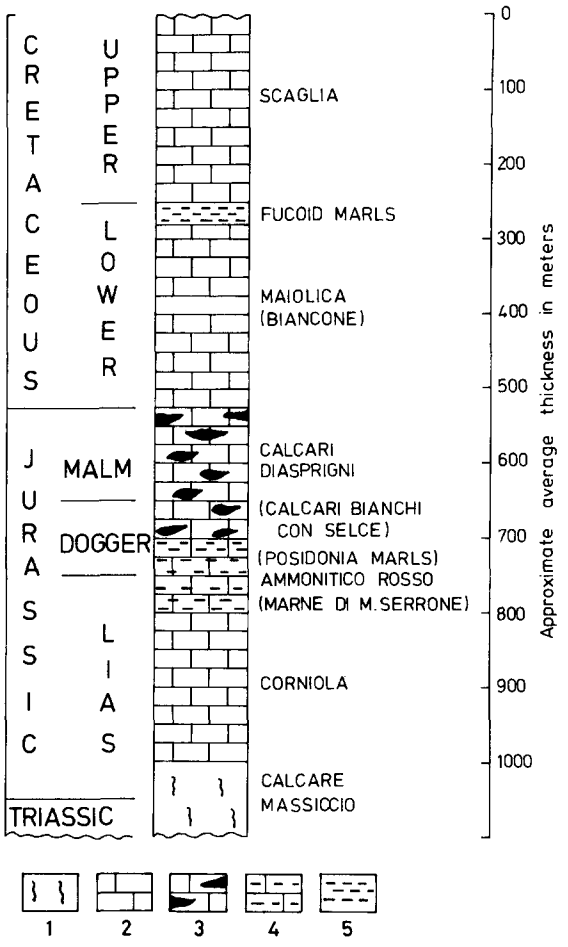


Fig. 1. Generalized stratigraphic column for the Late Triassic to Late Cretaceous time interval in the Umbrian Apennines. *1* = shallow-water platform limestones, *2* = pelagic limestones, *3* = pelagic cherty limestones, *4* = pelagic marly limestones, *5* = pelagic marls.

2. Stratigraphic sections

2.1. Valdorbia

The Valdorbia section (Fig. 2) is located on the western limb of the M. Cucco–M. Molette anticline, 1.5 km west of the small village of Valdorbia. This section is one of the most beautiful examples of a Jurassic pelagic sequence in the Tethyan realm. None of the formations has reduced thickness. There is no evidence, such as hard-grounds, diastems or paraconformities, for major gaps in sedimentation, although sedimentation rates may have been very variable. Slump structures are present in the upper part of the Ammonitico Rosso Formation.

The base of the sampled section (Fig. 3) is characterized by the Monte Serrone marls which are gray/green in color, with intercalations of marly calcareous turbidites. They are well dated by ammonites, as is the overlying Ammonitico Rosso Formation [7–9]. The lower part of the Ammonitico Rosso consists of red nodular limestones with intercalated red marls, and the upper part is mainly grey limestones separated by films of marl and clay. The transition between the upper and lower parts is gradational.

The transition between the Ammonitico Rosso and the overlying Calcari Diasprigni Formation consists of a sequence of 30 m of well bedded white limestones bearing grey nodules of chert,

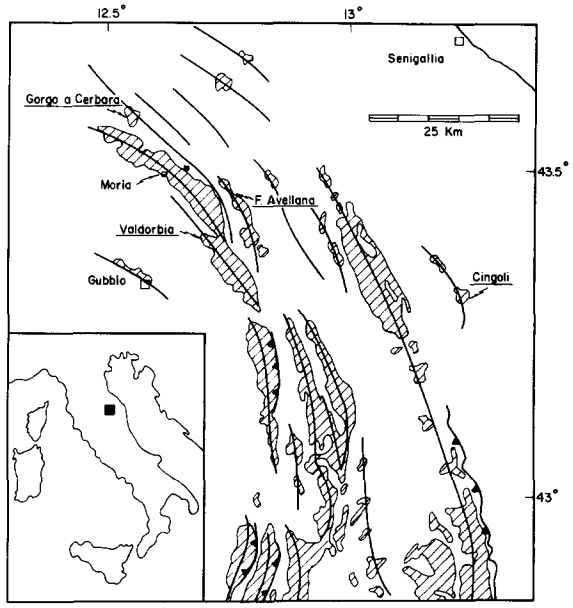


Fig. 2. Location map for the four Jurassic sections (underlined) relative to the Gubbio and Moria sections where the Late Cretaceous magnetostratigraphy was established. The shaded areas represent outcrop of the lower part of the stratigraphic section (Upper Triassic/Lower Cretaceous). The regional trend of anticlinal axes and the location of the more important thrust faults are shown.

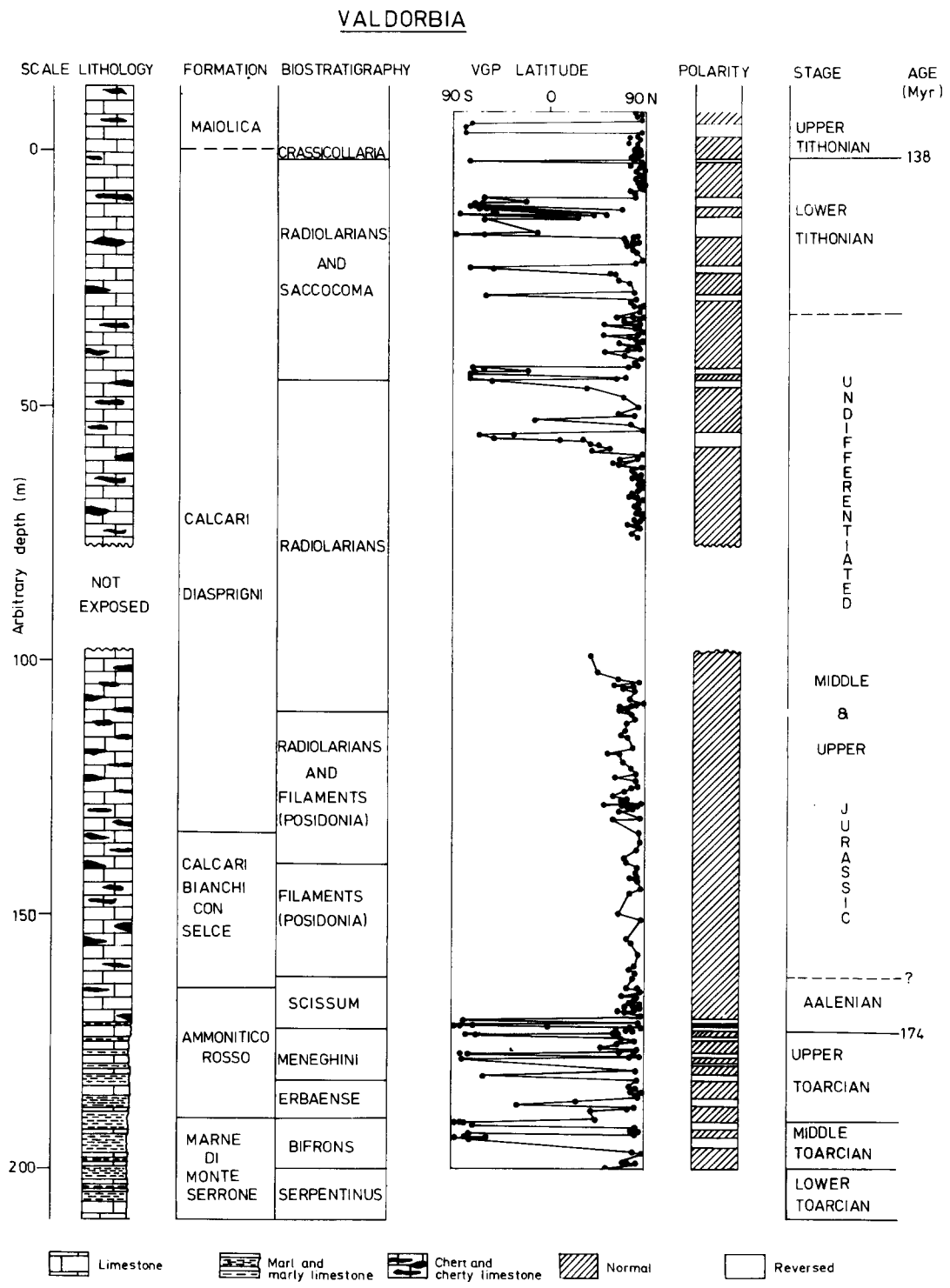


Fig. 3. The Valdorbia section.

known as the "Calcari Bianchi con Selce". Calcareous turbidites are scattered throughout the unit. The fossil content is very poor and does not give an exact indication of the age of the formation. "Filaments" (*Bositra* sp.) occur in the lower portion, and radiolarians and "filaments" in the upper part. The "Calcari Bianchi con Selce" is commonly assigned to the Dogger [10] with extension into the lower Oxfordian.

The Calcari Diasprigni is the thickest (110 m) formation outcropping in the section. It consists of thin, well bedded, turbiditic calcarenites and

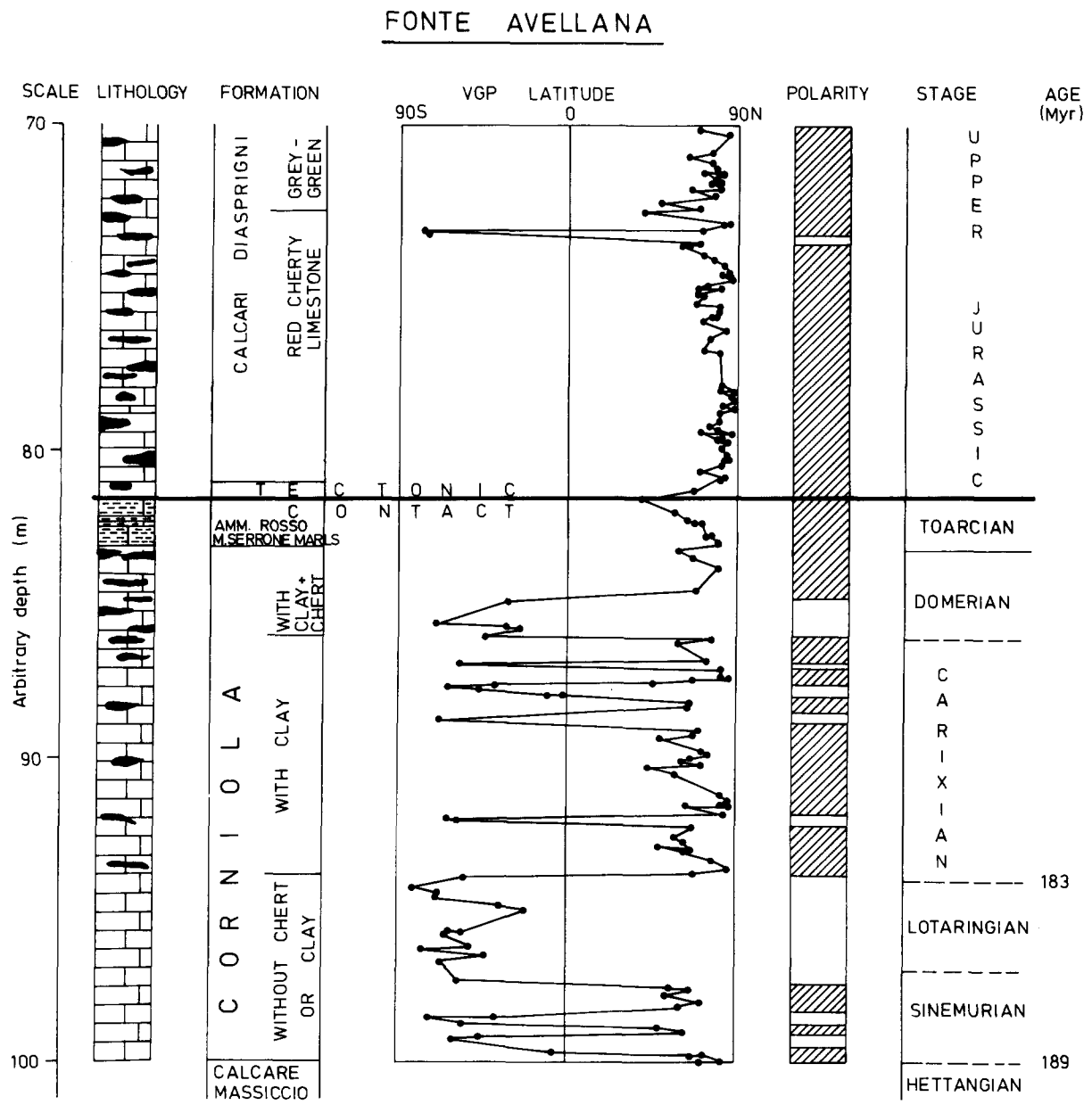


Fig. 4. The Fonte Avellana section.

hemipelagic mudstones, with interbedded layers and nodules of chert. The limestone/chert ratio is 4:1 at the top and bottom of the formation, and in the middle it decreases to 2.5:1. The color of the formation is light green/gray with a 2 m thick red interval in the middle. The contacts with the underlying "Calcari Bianchi con Selce" and with the overlying Maiolica are transitional. The fossil

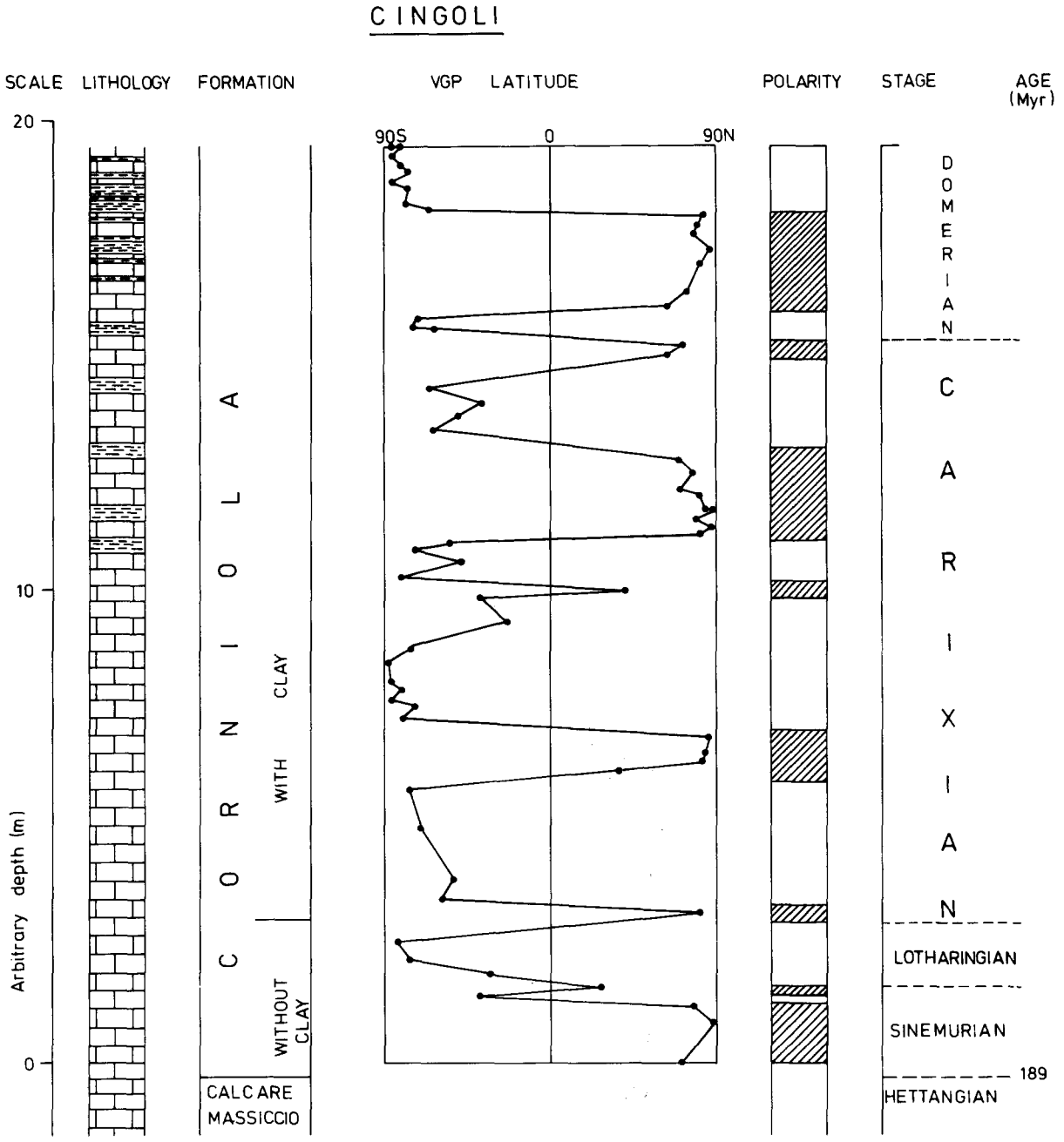


Fig. 5. The Cingoli section.

content of the Calcari Diasprigni consists mainly of Radiolaria and Aptyci (*Lamellaptychys beirichi*), and does not allow a good stage subdivision for the Late Jurassic. The transition to the overlying Maiolica bears a fauna belonging to the *Saccocoma and Stomiosphera* cenozone (assemblage zone) which can be assigned to the Lower Tithonian [11]. The overlying Maiolica is a white micritic limestone bearing thin layers of gray chert. *Crassicollaria* at the base of this formation give a late Tithonian age.

2.2. Fonte Avellana

A fine exposure of Jurassic rocks crops out along the Cesano river at Fonte Avellana (Fig. 2). Unfortunately a fault cuts off part of the stratigraphy, juxtaposing the M. Serrone marls with the Calcari Diasprigni (Fig. 4). The sequence is more condensed than that at Valdorbia but there are no apparent hiatuses in sedimentation. In most sequences in Umbria, the Calcare Massiccio/Corniola contact is either erosional or marked by a hard-ground, but here the contact is more transitional. The thick bedded grainstone-textured Calcare Massiccio with gastropods, pelecypods and crinoids grades into the thinner bedded Corniola with crinoids, brachiopods and pelagic organisms. The lower 7 m of the Corniola is characterized by stylolitic bedding planes, the middle 7.5 m by planar bedding surfaces marked by thin films of green clay, and the upper 2.5 m by the presence of chert nodules. The occurrence of the index fossil *Metaderoceras gemmellaroi* in the middle part of the formation gives a Carixian age. The M. Serrone marls overlie the Corniola but their thickness is reduced by a fault which brings the basal part of the M. Serrone marls into contact with the Calcari Diasprigni in which the very poor fossil content does not allow stratigraphic stages to be assigned.

2.3. Cingoli

The Cingoli section (Fig. 2) occurs in a quarry just west of a town of the same name, in the valley of the Rudielle river. The Corniola Formation at Cingoli (Fig. 5) is lithologically very similar to that at Fonte Avellana. The lower 6.5 m has stylolitic bedding surfaces and bed thickness decreases from 60 to 20 cm. The upper 13.5 m is characterized by films of clay separating thin, even bedding. Index fossils such as *Fucinicieras* sp. and *Protogrammoceras* sp. give a lower Domerian age for the uppermost part of the section. Assuming, as at Fonte Avellana, that the appearance of the first bed containing pelagic fauna above the Calcare Massiccio corresponds to the Hettangian/Sinemurian boundary and that the first appearance of clay marks the beginning of the Carixian (well documented at Fonte Avellana); then the lower part of the Corniola can be assigned to the Sinemurian and Lotharingian, the upper part to the Carixian, and only the uppermost beds to the lower Domerian.

2.4. Gorgo a Cerbara

The uppermost 15 m of the Calcari Diasprigni Formation have been sampled at Gorgo a Cerbara (Fig. 2) along the Candigliano river. The sampled interval comprises fine white calcarenites bearing gray chert nodules (Fig. 6). The fossil assemblage is characterized by Radiolaria and Saccocoma which give a lower Tithonian age.

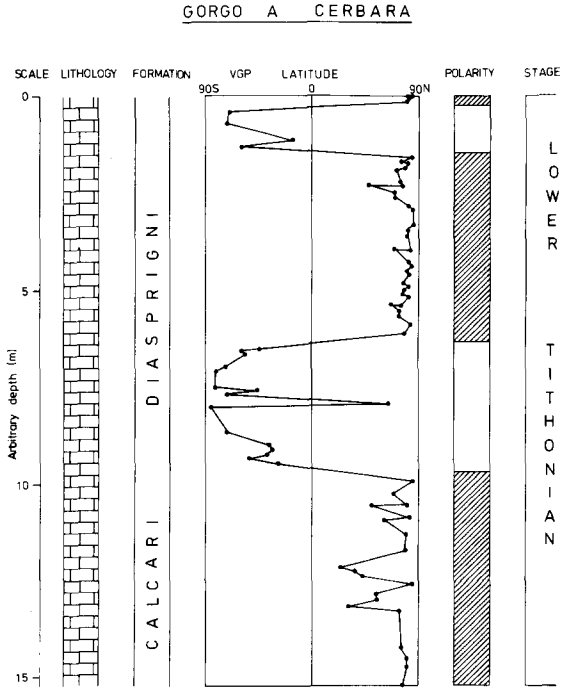


Fig. 6. The Gorgo a Cerbara section.

3. Magnetic properties

3.1. Red marls

The acquisition of isothermal remanent magnetization (IRM) is a useful guide to the coercivity spectrum of the magnetic minerals in a rock sample. IRM acquisition curves from the Ammonitico Rosso (Fig. 7) indicate an initial rapid increase of IRM in magnetizing fields up to 0.5 T, followed by a slow approach to saturation IRM which is acquired in magnetizing fields of about 1 T. The

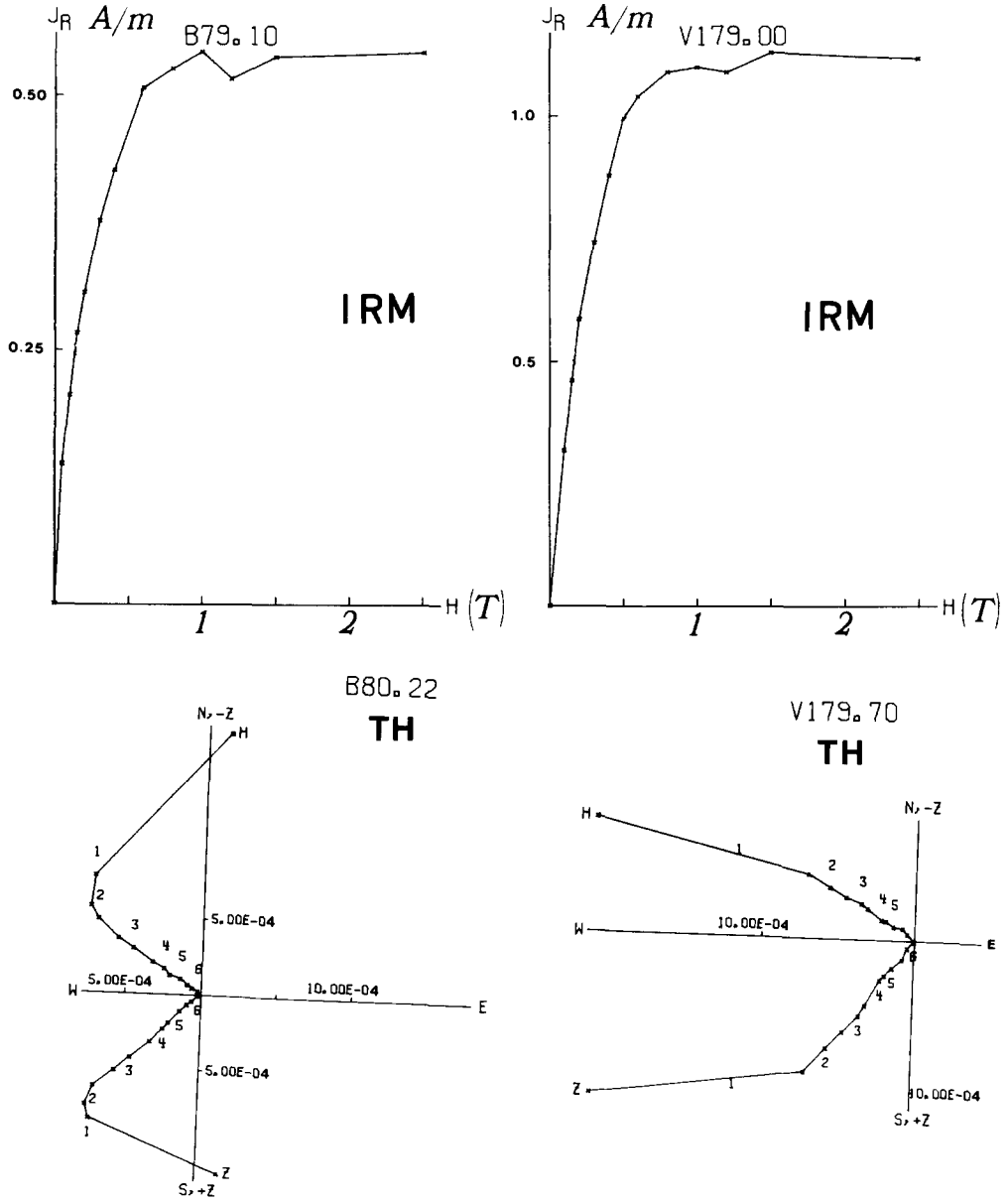


Fig. 7. IRM acquisition curves, and orthogonal projections of thermal demagnetization data, for samples of Ammonitico Rosso from Fonte Avellana (left) and Valdorbia (right). The horizontal (H) and vertical projections (Z) for the orthogonal projections are indicated as are the demagnetization temperatures ($\times 10^2$°C). Magnetization intensities are given in A/m.

rapid initial rise is due to low-coercivity magnetite, whereas the high coercivities are undoubtedly due to the fine-grained haematite which gives these rocks their red hue. Although haematite is a major contributor to the saturation IRM, its contribution to the natural remanent magnetization (NRM) is less clear. The NRM is completely destroyed by demagnetization at temperatures over 575°C (Fig. 7); and the vector decay to the origin is linear above 200°C. If both haematite and magnetite contribute after demagnetization at temperatures above 200°C, their blocking temperature spectra exactly coincide or their directions are parallel (or antiparallel). As the fine-grained haematite pigment might be expected to have high coercivity but low blocking temperature, thermal demagnetization may have successfully removed the haematite contribution at temperatures above 200°C. The pigmentary haematite in these rocks undoubtedly grows during diagenesis although rock magnetic studies in Cretaceous Scaglia Rossa [12] indicate that the largest grains of pigmentary haematite (with higher blocking temperature) grow through their critical volume very soon after deposition. Therefore, although we do not have a clear picture of the relative contributions of haematite and magnetite to the NRM, we can consider the higher blocking temperature magnetizations to be recording the geomagnetic field at a time contemporaneous (geologically speaking) with deposition.

3.2. White limestones

Whereas haematite is an important contributor to the magnetization of the red marls, the magnetization of the white limestone formations (Calcari Diasprigni and Corniola) is due largely to magnetite. The IRM usually reaches saturation in magnetizing fields as low as 100 mT (Fig. 8), indicating the absence of high coercivity minerals, and alternating fields are usually an efficient means of demagnetizing the NRM (Fig. 9). However, for the Valdorbia section, the Calcari Diasprigni and "Calcari Bianchi con Selce" limestones were also treated thermally. Examples of thermal demagnetization data for various meter levels within the prolonged interval of normal polarity (Fig. 10) usually indicate univectorial decay, although the magnetization often falls below the noise level of the magnetometer at demagnetization temperatures above 400°C. The fact that the single component survives demagnetization temperatures above 350°C indicates that it cannot be viscous in origin [13]. Indeed when we plot the magnetization directions before and after bedding correction for the entire Valdorbia section (Fig. 11), the characteristic magnetization directions before bedding correction are significantly different from the present Earth's field, and the directions after bedding correction (Table 1) for Valdorbia, as well as for the other sampled sections, are close to those previously obtained for the Umbrian Jurassic [14]. The mean inclinations are not significantly different from other northern Umbrian Jurassic or pre-Santonian Cretaceous data. The declinations are occa-

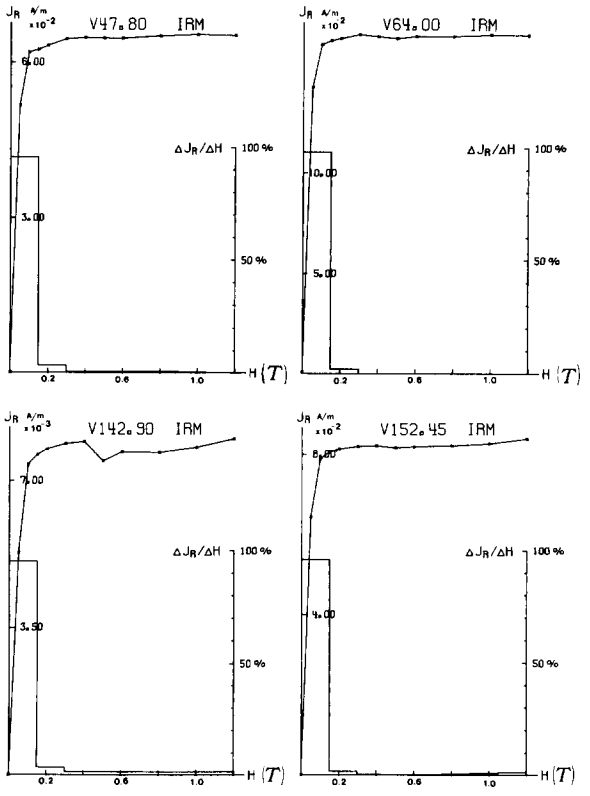


Fig. 8. IRM acquisition curves for Calcari Diasprigni samples from Valdorbia. The block diagram beneath the acquisition curve indicates the change of IRM intensity (J_R) over various increments of applied field (H) expressed as a percentage of the saturation magnetization.

sionally significantly different from site to site. This has been previously noted in Umbrian Cretaceous data and has been explained in terms of the deformation style in Umbria [15]. However, some of the directional scatter is reflected in imperfect antiparallelism of normal and reversed directions (Fig. 11) which suggests that thermal cleaning was not always complete. Extensive thermal and alternating field demagnetization of samples from the Valdorbia section has failed to reveal any samples with reversed magnetization in the middle part of the section (Fig. 3). The low coercivity and blocking temperatures of the Calcari Diasprigni and "Calcari Bianchi con Selce" magnetizations tend

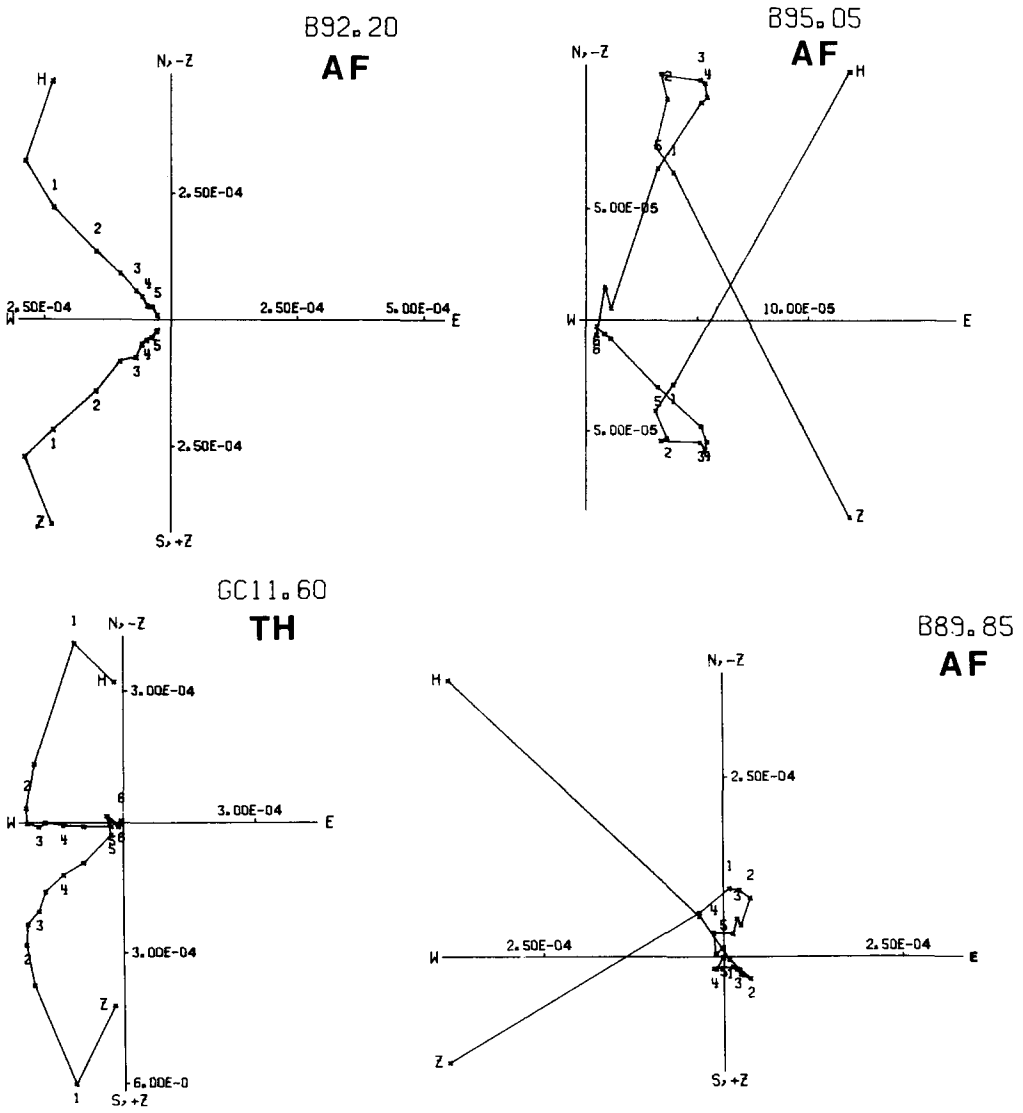


Fig. 9. Orthogonal projections of thermal (TH) and alternating field (AF) demagnetization data for samples of Calcari Diasprigni from Fonte Avellana (*B*) and Gorgo a Cerbara (*GC*). The horizontal (*H*) and vertical (*Z*) projections are indicated. The peak demagnetization fields (×10 mT) and the demagnetization temperatures ($\times 10^2$°C) are given. Magnetization intensities are in A/m.

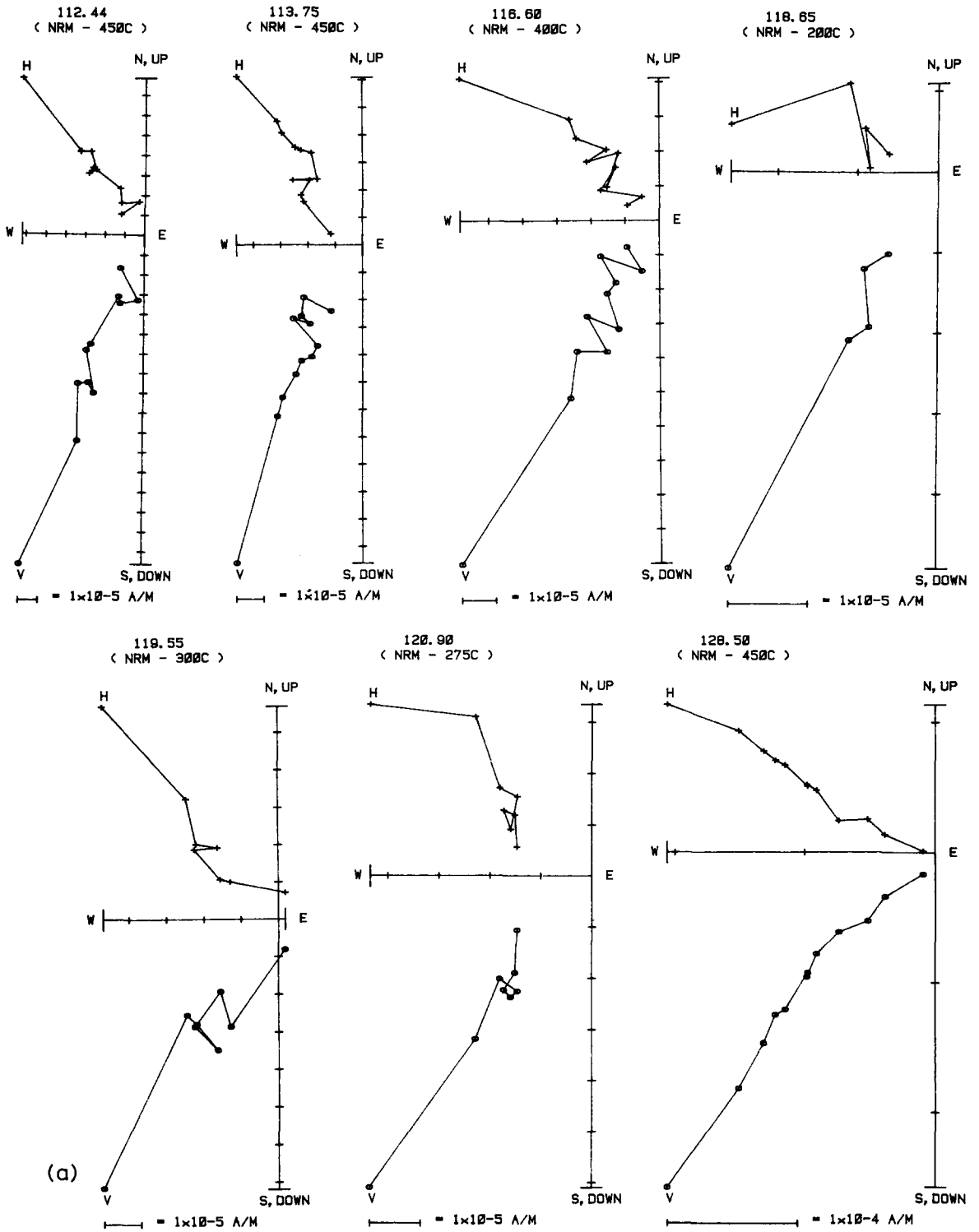
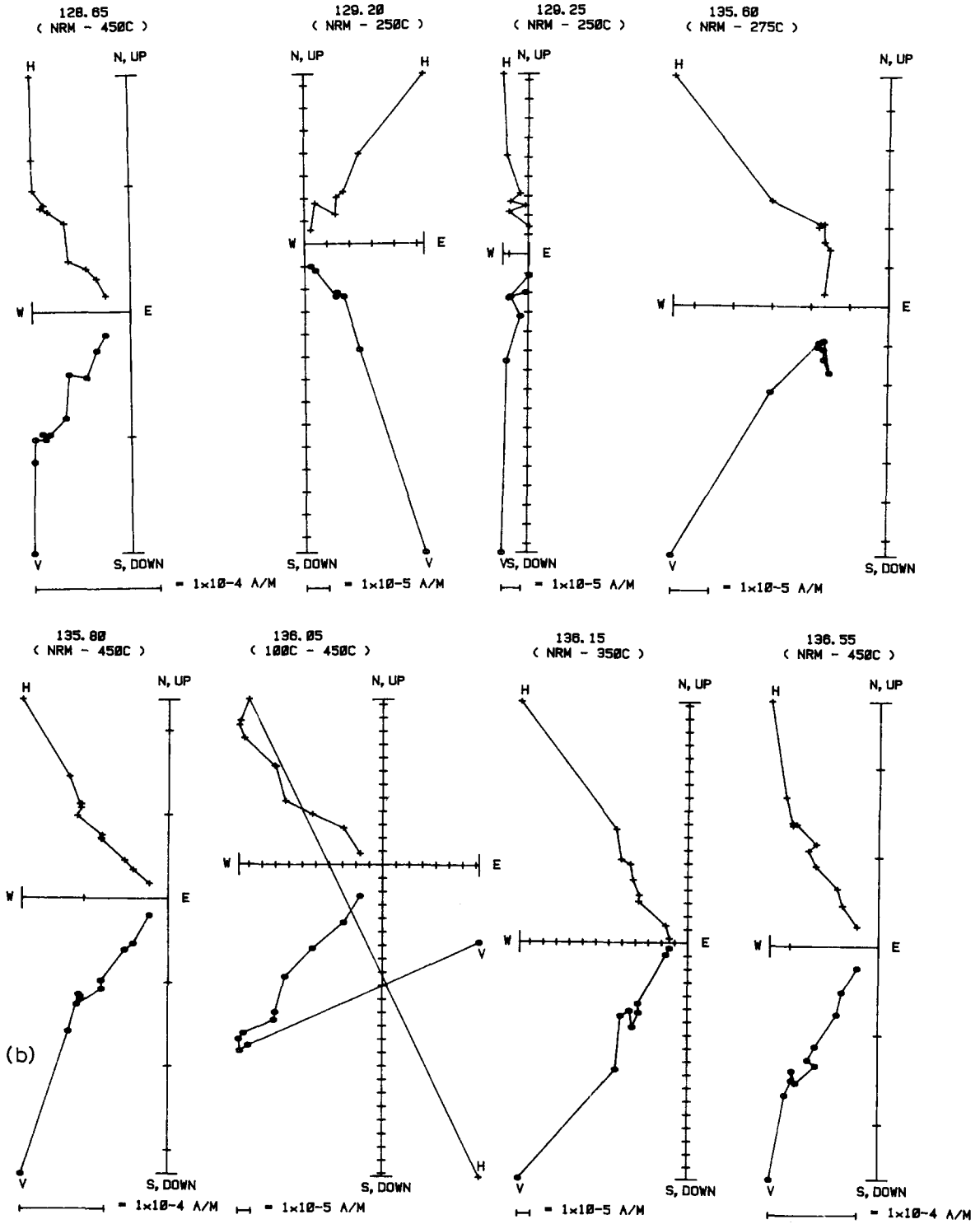


Fig. 10. (a, b) Orthogonal projections of thermal (TH) demagnetization data of samples from various meter levels within the interval of prolonged normal polarity at Valdorbia. The meter levels and the range of demagnetization temperatures ($\times 10^2$°C) are given.



Crosses and circles represent projection on the horizontal and vertical planes respectively. The noise level of the magnetometer was such that samples with intensities less than 1×10^{-5} A/m could not be precisely measured.

to suggest that magnetite is the principle remanence carrier. If the magnetite is detrital or biological in origin, its remanence in a rock sample is the result of mechanical alignment in the soft sediment, shortly after deposition [16,17]. Magnetite, however, can be authigenic and magnetite

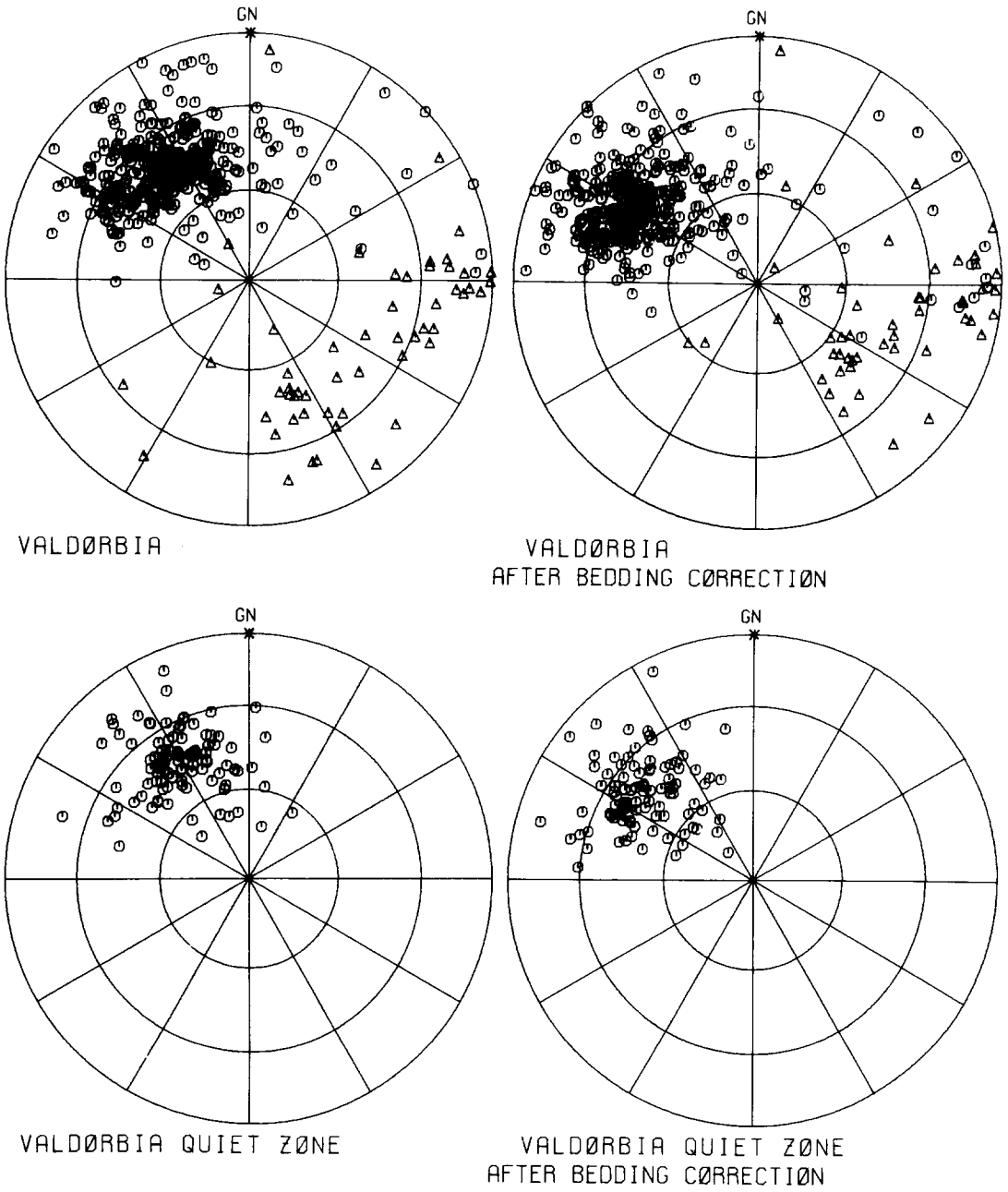


Fig. 11. Stereographic projection of magnetization directions from Valdorbia after thermal cleaning, for the whole section (above) and for the "quiet-zone" (below).

TABLE 1

Characteristic mean magnetization directions, after tilt correction, for various subdivisions of the sampled sections

Section	N	$D(°)$	$I(°)$	$\alpha_{95}(°)$	K	VGP	
						lat.	long.
Valdorbia (complete)	351	299.7	38.2	2.5	10.1	35.9	279.1
Valdorbia (upper mixed)	163	292.7	31.3	3.9	9.2	27.9	279.9
Valdorbia (quiet zone)	95	308.1	41.8	3.2	22.2	43.6	275.5
Valdorbia (lower mixed)	93	303.3	46.5	5.0	9.6	42.3	283.1
F. Avellana (complete)	176	310.2	41.4	4.2	7.4	44.9	273.4
F. Avellana (above fault)	82	312.1	40.7	4.3	14.3	45.9	271.3
F. Avellana (below fault)	94	308.4	42.1	7.1	5.2	43.9	275.4
Cingoli	68	336.0	55.3	5.6	10.3	70.2	269.2
Gorgo a Cerbara	75	259.2	37.7	7.3	6.4	7.0	305.1

of diagenetic origin has been recognized in North American shallow-water marine limestones [18]. The folding in Umbria is Miocene/Pliocene in age, and therefore a fold test is very little use in confining the age of the magnetization. As the magnetization directions are indistinguishable from middle Cretaceous to Jurassic Umbrian data, remagnetization by growth of an authigenic magnetic phase, such as magnetite, during the *Cretaceous* quiet-zone cannot be ruled out.

4. Magnetic stratigraphy

The samples were measured on a cryogenic magnetometer capable of measuring the magnetization along three orthogonal directions at the same time. The magnetization of each sample was measured in three different configurations relative to the magnetometer axes to control the measurement repeatability [17]. If the circular standard deviation (ψ) for the three independent measurements was excessive, the measurement was rejected:

$$\psi = 81/\sqrt{k} \text{ degrees}, \quad k = (N-1)/(N-R)$$

where N is the number of independent measurements ($N = 3$) and R is the length of the resultant vector giving unit weight to each measurement.

The virtual geomagnetic polar (VGP) latitude is used as the measure of polarity in Figs. 3–6. The VGP latitudes are computed for each individual sample and represent the latitudes relative to the palaeomagnetic pole computed from the mean direction for the whole section.

The section at Valdorbia is the longest and most complete of the sampled sections. The interval sampled was from Lower Toarcian to Upper Tithonian (Fig. 3). Samples were collected at roughly half-meter intervals throughout the 200-m section. The VGP latitudes plotted in Fig. 6 are derived from the remanent magnetization directions after thermal demagnetization at 350°C with some supplementary AF data from the Calcari Diasprigni and "Calcari Bianchi con Selce" Formations. As discussed above, alternating field demagnetization is an efficient technique for isolating the characteristic remanence of the Calcari Diasprigni, but is inadequate in the case of the Ammonitico Rosso Formation. The predominant feature of the polarity stratigraphy is the existence of two mixed polarity intervals separated by an extended period of normal polarity, which is probably the equivalent of the Jurassic"quiet-zone" in

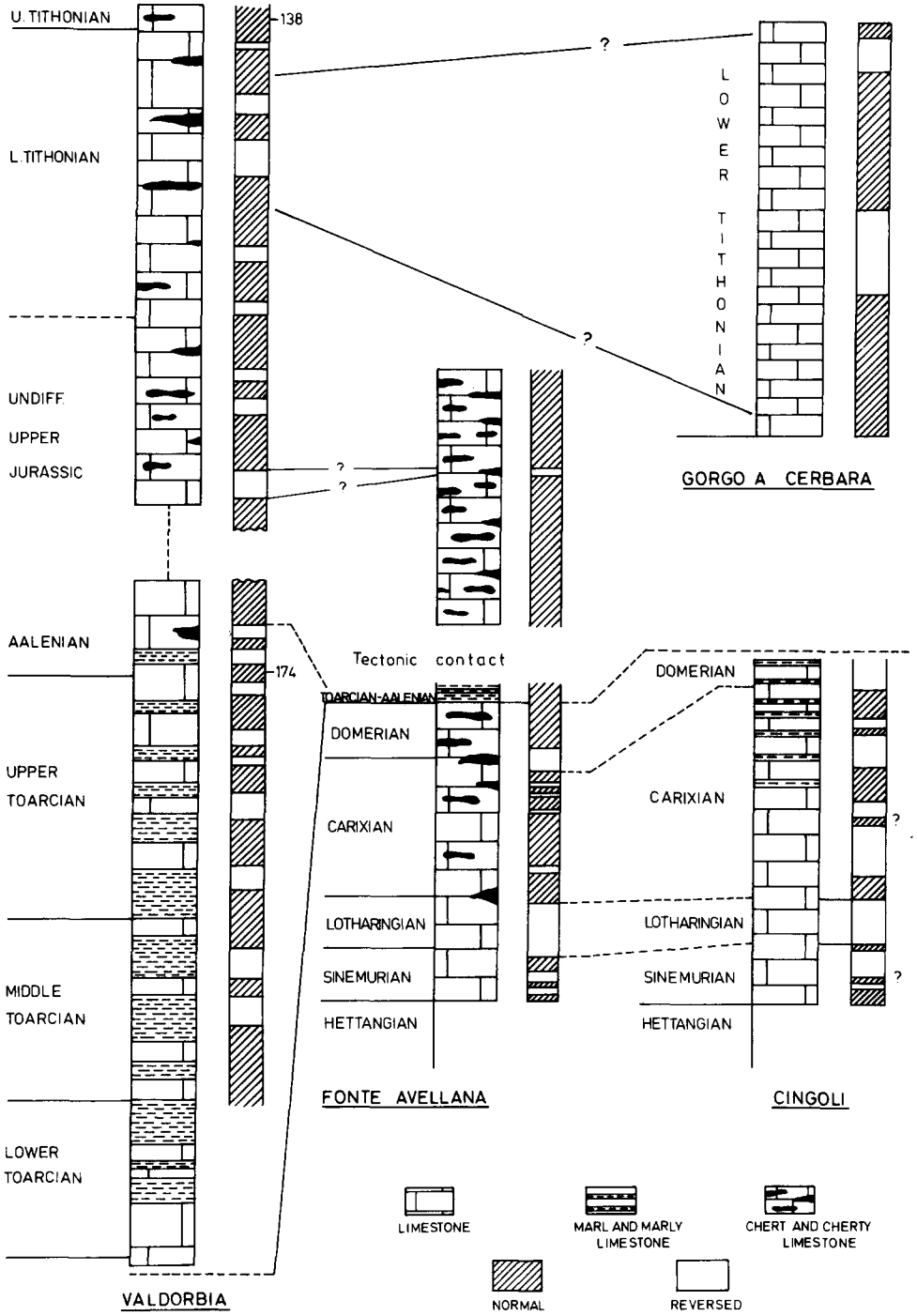


Fig. 12. Tentative correlation of the mixed polarity intervals from the various sections based on the palaeontological, lithological and magnetostratigraphic data.

the oceanic magnetic anomaly record. The "quiet-zone" magnetization directions are similar to those in the remainder of the section (Fig. 11).

The VGP latitudes for the Fonte Avellana section (Fig. 4) are derived from the magnetization directions after thermal demagnetization at 300°C, although alternating field treatment would have been sufficient for both the Calcari Diasprigni and Corniola. As in the Valdorbia section (Fig. 3), there is a stratigraphic interval of predominantly normal polarity. Most of the Lias is clearly of mixed polarity. Unfortunately, the palaeontological control is again poor.

At Cingoli, palaeontological control is again not good; however, the VGP latitudes (after thermal demagnetization at 300°C) indicate mixed polarity for this interval (Fig. 5). At Gorgo a Cerbara (Fig. 6), two reversed intervals are apparent after thermal demagnetization. The correlation with the Valdorbia section is ambiguous without further paleontological data.

5. Discussion

The Umbrian magnetostratigraphic data support the conclusion that the Jurassic period can be divided into three intervals on the basis of geomagnetic polarity. Two intervals of mixed polarity are separated by an interval of quasi-constant normal polarity. We suppose that the younger mixed interval corresponds to the clearly defined lineated oceanic anomalies at the old end of the M-sequence [20]. The older mixed interval corresponds to a time of alternating magnetic polarity early in the Jurassic, during and prior to the break-up of Pangea, for which there is no clear expression in the oceanic magnetic anomaly record. The interval of quasi-constant normal polarity, one reversal recorded at Fonte Avellana, may correspond to the Jurassic magnetic smooth zones in which lineated oceanic anomalies are absent or poorly defined (i.e. in the western central Atlantic between M-25 and the Blake Spur anomaly [21]). Unfortunately, poor paleontological control does not allow either a precise estimate of the duration of this interval or an unambiguous correlation between mixed polarity intervals from different sections (Fig. 12). The variable abundance of turbidites in the Valdorbia section suggests fluctuations in sedimentation rate, and as the Calcari Diasprigni and "Calcari Bianchi con Selce" Formations contain abundant turbiditic horizons, their sedimentation rates might be expected to be higher than those of formations above and below. Indeed, magnetostratigraphic constraints on the age of the "quiet-zone" from the southern Alps [22], where a mixed polarity section of Bajocian/Bathonian age has been studied and M-25 has been extrapolated into the Oxfordian, confine the prolonged normal polarity interval to the Callovian and Oxfordian stages. Within the paleontological resolution of stage boundaries, this is consistent with studies of shallow-water limestones from southern Germany [23,24] where Oxfordian and early Kimmeridgian rocks are normally magnetized. An aptychus assemblage suggests a Callovian age for the onset of radiolarite

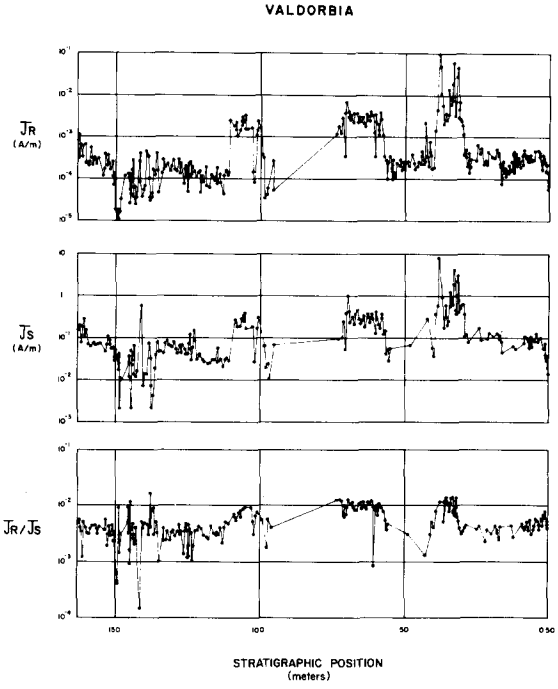


Fig. 13. Variations in the intensity of the NRM (J_R) and of the saturation IRM (J_S) plotted on a logarithmic scale as a function of stratigraphic position at Valdorbia (Fig. 3). The ratio of NRM to saturation IRM, plotted on a logarithmic scale, does not show a clear gradual change which might indicate a systematic change in geomagnetic field intensity.

(Diaspri) sedimentation in southeastern Tuscany [25], but the correlation between these pelagic rocks and those in Umbria is poorly understood.

The basement in the western Atlantic Jurassic quiet-zone at DSDP Site 534 was found, from oldest overlying sediments, to be middle Callovian in age [26]. The site is situated about 400 km landward of M-25 and about 100 km east of the Blake Spur anomaly, its location and basement age are consistent with the hypothesis that the "quiet-zone" is not older than Callovian.

Steiner [27] has reviewed the palaeomagnetic data from a number of middle and late Jurassic formations in Colorado, Utah and Wyoming. The magnetization of these sediments is usually very complex and a period of extended normal polarity in the middle and late Jurassic is certainly not apparent from the data. However, the quality of the data is such that the possibility cannot be excluded [28].

Cande et al. [29] have suggested that the Jurassic "quiet-zone" in the oceanic magnetic anomaly data is not the result of a prolonged period of normal polarity but due to a reduction in the intensity of the geomagnetic field at that time. In order to test this suggestion we have measured the value of saturation IRM for samples from the upper 163 m of the Valdorbia section (Fig. 3). This value would be expected to vary with magnetite content in the Calcari Diasprigni Formation. Provided the grain size spectrum of the magnetite is uniform throughout the formation, we can use the saturation IRM as a means of normalizing the NRM for variations in magnetite content, and hence obtain a means of monitoring changes of geomagnetic field strength during deposition of the formation. The NRM fluctuations are closely mirrored by fluctuations in saturation IRM (Fig. 13). The ratio of NRM to saturation IRM does not show a clear trend which might indicate a systematic reduction of geomagnetic field strength during middle and late Jurassic.

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